

Work In Progress: Towards Design Criteria for Virtual Humans' Bodily Emotional Gestures in VR Public Speaking Experiences

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ABSTRACT

We propose the initial design criteria for the bodily emotional gestures (BEG) of virtual human audiences (VHs) during presentation in a virtual reality (VR) public speaking experiences. Guided by design criteria proposed in earlier studies (approach–avoidance, engaged–distracted, and prototypical audience behaviors), we developed a set of gestures intended to elicit emotions across the four quadrants of the valence–arousal space. Following an initial validation through an online survey ($N = 126$), we identified five emotional bodily behaviors, including a neutral condition, as the most promising for eliciting quadrant-specific emotional responses for future formal evaluation. Based on these findings, this paper discusses the technical details of the BEGs, the next steps for their evaluation, and potential implications.

Index Terms: Virtual Audience, Bodily Emotional Gesture, Public Speaking, Perceptual Response, Emotion

Index Terms: Human-centered computing—Virtual reality; Computing methodologies—Perception; Human-centered computing—User studies; Human-centered computing—User interface design;

1 INTRODUCTION

In the context of public speaking, real human audiences evoke a wide range of emotional and behavioral responses in speakers, including heightened arousal and sensitivity to feedback [1, 6, 9, 20]. Similarly, virtual human audiences (VHs) in immersive virtual reality (VR) have been shown to elicit comparable effects: speaking in front of a VH can induce measurable anxiety and modulate self-reported stress, physiological reactivity, and performance [3, 5, 21, 25]. These properties make VHs valuable tools for a range of VR applications, including public-speaking training, education, and therapeutic contexts. In public-speaking training, supportive VH behaviors (e.g., nodding or maintaining eye contact) can facilitate the rehearsal of speaking skills [22]. VH animations have also been shown to elicit distinct cognitive and attentional responses, reflected in eye-movement and visual-search measures [11] in VR classroom environment. Beyond training, VHs have been applied in collaborative VR environments, where avatar-based nonverbal feedback enhances engagement and social presence [12]. In exposure-based therapy for social anxiety disorders, VHs provide a controllable, graduated, and safe social environment in which users can confront

anxiety-inducing situations while receiving real-time social feedback [3]. Across these domains, the design of bodily expressive behaviors in VHs is critical, as such behaviors shape users' emotional responses and directly influence the effectiveness of VR-based interventions.

Despite the increasing adoption of emotional avatars, the design of avatar body gestures remains heavily dependent on individual designers' intuition. To the best of our knowledge, relatively little work has established standardized design criteria for bodily emotional gestures in VR. Moreover, the role of communication context has not been adequately addressed, despite its importance for gesture design. For example, bodily emotional gestures in educational settings, such as public presentation in classroom, differ from those in everyday communication: educational gestures tend to be more purposeful and functional, supporting comprehension and shaping the learning environment. These differences highlight the need for systematically designed bodily emotional gestures, particularly in educational and public-speaking contexts.

To address this gap, as a preliminary investigative step, we propose a set of design criteria for virtual humans' bodily emotional gestures (BEGs) in public-speaking scenarios involving groups of virtual human audiences. These BEGs are grounded in the Valence–Arousal emotion model [4, 19] and the Approach–Avoidance framework, in which posture and body orientation serve as key parameters in gesture design [7, 31, 32]. Together, these frameworks provide systematic mappings between audience behaviors and affective dimensions, for example, nodding corresponding to positive, high-arousal states, which are discussed in more detail in the following section.

2 RELATED WORK

2.1 Impact of Emotional Virtual Human

Prior research has shown that virtual humans [2, 18, 23, 30] can significantly influence users' emotions [33, 14], cognition [27, 35], subjective experiences [26, 17], and task performance [15, 29, 34]. Early studies demonstrated that virtual human audiences (VHs) can evoke realistic stress responses and public speaking anxiety [25]. Both explicit verbal behaviors and behavioral reactions of avatars have been shown to influence users' emotional states [14, 33], as well as their cognitive processing, experiences, and task performance [26, 27, 29, 34]. Beyond verbal cues, nonverbal avatar responses, such as posture, gesture, and body orientation, also play a significant role in shaping users' emotional responses [14, 33]. Extending these findings, recent research suggests that the emotional impact of virtual audiences' bodily expressive behaviors is amplified as the number of avatars increases, highlighting the importance of group-level embodiment and coordinated expressivity in immersive environments [16]. Collectively, these findings underscore the importance of not only the presence of virtual audiences but also the careful design and coordination of their behaviors.

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2.2 Valence-Arousal Model of Emotion

The Valence-Arousal model [4, 19] provides a dimensional framework that represents human emotions along two continuous axes: valence (positive to negative) and arousal (high to low). This model is commonly used in affective computing and emotion-driven interaction design [13, 24] to assess whether a given stimulus, such as clapping of a VH, elicits the intended emotional response (e.g., a strong positive emotion). In this model, valence essentially captures the hedonic tone of an emotion, ranging from unpleasant (negative) to pleasant (positive). Emotions with positive valence include joy, excitement, or contentment, while emotions with negative valence include sadness, anger, or fear. Arousal represents the level of physiological or psychological activation associated with an emotion, ranging from low to high. Emotions with high arousal include surprise, anger, or enthusiasm, involving increased alertness, heightened attention, or bodily activation. In contrast, low-arousal emotions include calmness, boredom, or melancholy, associated with reduced activation or passivity. Together, these two dimensions provide a continuous and interpretable representation of a broad spectrum of emotions.

2.3 Approach-Avoidance Criterion

One of the design criteria for VH is the Approach-Avoidance framework from the field of psychology. This framework associates approach-related behaviors with positive valence and high arousal, and avoidance-related behaviors with negative valence and low arousal [8, 10]. According to this framework, postures and orientations are considered as useful parameters in gesture design. For example, gestures featuring forward-leaning postures, open chest orientation, and direct body orientation toward the user would convey engagement and positive affect [31, 32]. Conversely, slouched postures, closed body language, and averted torso or head orientation would express disinterest or negative affect [7].

3 BEG DESIGN CRITERIA

We heuristically applied design criteria adopted by previous researchers (e.g., avoid-approach, eye gaze, prototypical behaviors) to the VHs seated in desk-chair configurations in our virtual classroom context. Accordingly, we selected audience reactions that could be expressed primarily through torso and arm movements. For instance, to design gestures conveying positive emotions, we drew on the avoid-approach criteria and eye-gaze behaviors, making the avatars lean forward and maintain eye contact. Conversely, for negative emotions, we designed the avatars to lean backward and avert eye contact. To manipulate arousal, we varied the expressiveness of the arms (e.g., crossed arms for low arousal and open arms for high arousal), consistent with prototypical behaviors identified in prior work. Through this design process, we generated three gestures for each of the four quadrants in the valence-arousal model (Figure 1), which are listed below.

Negative Valence We selected the following six behaviors indicating disengagement or disagreement, which were assumed to elicit negative valence from the speaker: raising both hands in the air aggressively in disagreement or disappointment (A), covering heads with both hands (B), resting the chin on a hand without making eye contact (C), dozing off (G), leaning back with both hands behind the head (H), and texting (I).

Positive Valence We selected the following six reactions indicating engagement and agreement that would yield positive valence: scratches head while taking notes, occasionally glancing at the presenter. (D), taking notes with occasional eye contact (E), applauding politely (F), looking forward with a hand on the chin (J), and looking forward with both hands on the desk (K) or just below the chin (L).

Table 1: Demographics result in the initial survey

	Male	Female
# of Participants	53	72
M (SD)	26.94 (10.29)	21.19 (5.57)
Asian / Pacific Islander	6	12
Black / African American	10	17
Caribbean	0	1
Hispanic / Latino	8	10
Middle Eastern	1	1
Puerto Rican and Jamaican	0	1
White / Caucasian	29	29
Prefer not to say	0	1

3.1 Survey

Before conducting the formal user study to evaluate BEG in an immersive setting, we first collected initial perceptual impressions of each gesture through a web-based survey. This preliminary study aimed to identify the most appropriate gesture cases for the user study, rather than evaluating all possible conditions. The survey was distributed internationally through academic communities in 3D user interfaces. In the survey, participants viewed each avatar's body gesture animation on their monitors or mobile devices. They then rated their emotional impressions of each gesture in terms of valence and arousal, using a 0 to 10 Likert scale. We targeted a sizable sample (over 100 participants) to minimize bias stemming from differences in the display medium compared to the immersive environment used in the main study. Since the terms "Valence" and "Arousal" may not be familiar to the general public, we replaced "Valence" with "Mood" (0: Extremely Negative, 5: Neutral, 10: Extremely Positive) and "Arousal" with "Intensity" (0: Not Intensive At All, 5: Neutral, 10: Extremely High Intensity). We also clarified that the avatars did not display facial expressions, instructing participants to focus solely on the body gestures within the classroom context. The study, including this initial survey, was approved by our institution's IRB, and participants provided informed consent before voluntarily participating.

3.2 Result

We collected a total of 126 survey responses, including 53 males (M = 26.94, SD = 10.29), 72 females (M = 21.19, SD = 5.57), and 1 "Not to prefer" response (see Table 1). We averaged the participants' responses after converting them to a -5 to 5 Likert scale, as this effectively mapped the responses to the valence-arousal quadrants. We then calculated the euclidean distance (ED) using Mood (M) and Intensity (Int), represented by the formula $\sqrt{Mood^2 + Intensity^2}$. The most extreme cases from each quadrant, identified as the gestures farthest from the center (0,0), were selected for the future user study. We also selected a neutral gesture D with the closest distance to the center. Accordingly, we mapped the following gestures: Gesture F (Mood: 2.93 (3.02) Int: 1.56 (3.24), ED: 3.32) to the first quadrant (Q1), Gesture A (Mood: -2.87 (2.91) Int: 2.43 (3.37), ED: 3.76) to the second quadrant (Q2), Gesture G (Mood: -2.23 (2.93) Int: -0.33 (3.82), ED: 2.25) to the third quadrant (Q3), Gesture H (Mood: 1.35 (3.82) Int: -1 (3.31), ED: 1.68) to the fourth quadrant (Q4), and Gesture D (Mood: 0.13 (2.81) Int: 0.19 (3.2), ED: 0.23) as neutral (N) (see Figure 2). Note: Based on their ED in the valence-arousal space, H was chosen for quadrant 4.

4 DISCUSSION AND FUTURE WORK

The survey results support the general validation of the BEGs, however, further evaluation in immersive settings is needed. In particular, the discrepancy observed for Gesture H may stem from the display medium, potential cultural differences, ironic interpretations [28], or interpretive bias among international survey partic-

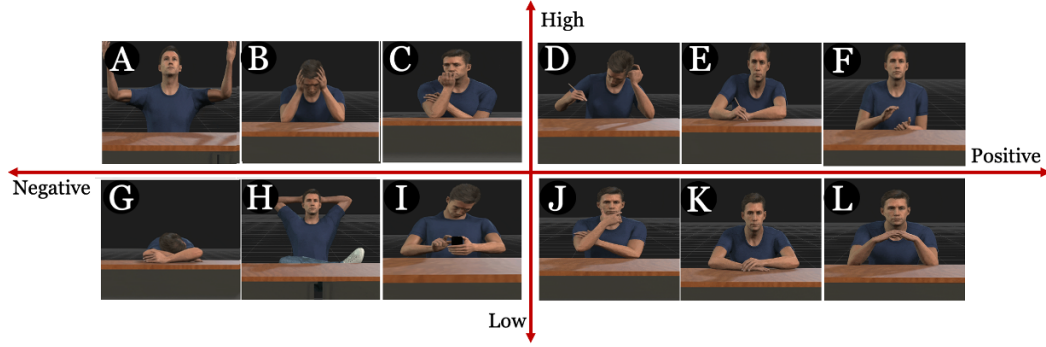


Figure 1: Designed body gestures based on BEG: Aggressive (discouraging) or Engaged (encouraging) gestures are related to the Valence, with wideness or active of arm gestures. Active (or agitated) or Passive (or calm) gesture in term of engagement to the activity related to the Arousal. A: Raising both hands in the air aggressively in disagreement or disappointment. B: Covering heads with both hands. C: Resting the chin on a hand without making eye contact. D: Scratches head while taking notes, glancing at the presenter. E: Taking notes with occasional eye contact. F: Applauding politely. G: Dozing off. H: Leaning back with both hands behind the head. I: Texting. J: Looking forward with a hand on the chin. K: Looking forward with both hands on the desk. L: Both hands below the chin.

Table 2: Results from the evaluation of the BEG in the initial survey. The ID refers to the designed gestures (A to L), DVs represent the responses, and M(SD) indicates the descriptive statistics. Q denotes the quadrant, and ED represents the Euclidean distance. Based on the ED, we selected gestures F, A, G, and H (instead of J). **Bold** highlights the selected gestures for the user study.

ID	DVs	M (SD)	Q	ED	ID	DVs	M (SD)	Q	ED
A	Mood	-2.24 (-2.67)	2	2.62	D	Mood	-0.24 (-2.81)	1	1.07
	Intensity	1.29 (-2.16)				Intensity	-0.21 (-2.34)		
B	Mood	-2.26 (-2.76)	2	2.4	E	Mood	0.52 (-2.45)	1	2.25
	Intensity	0.24 (-2.1)				Intensity	1.16 (-2.24)		
C	Mood	-1.33 (-2.54)	2	1.65	F	Mood	1.62 (-2.54)	1	2.39
	Intensity	0.65 (-2.08)				Intensity	0.74 (-2.34)		
G	Mood	-1.82 (-2.65)	3	3.04	J	Mood	0.38 (-2.69)	4	1.54
	Intensity	-0.56 (-1.74)				Intensity	-0.17 (-2.2)		
K	Mood	-0.71 (-2.84)	3	1.51	H	Mood	0.6 (-2.2)	4	1.3
	Intensity	-1.25 (-1.99)				Intensity	-0.98 (-2.24)		
I	Mood	-0.69 (-2.33)	3	1.58	L	Mood	0.1 (-2.56)	4	1.29
	Intensity	-0.37 (-1.91)				Intensity	-0.27 (-2.21)		

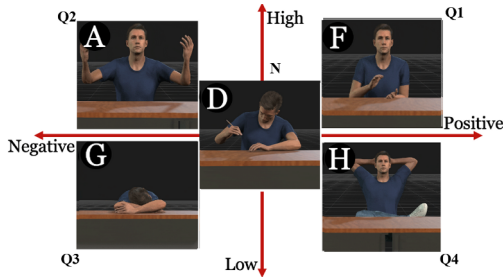


Figure 2: Selected body gestures from the survey. The most extreme cases with the farthest Euclidean distance from the center of the quadrant (0,0) were selected for each quadrant in the formal user study. Gesture F was mapped to Q1, gesture A to Q2, gesture G to Q3, gesture H to Q4, and gesture D as Neutral (with the closest Euclidean distance from the center).

ipants. These factors warrant additional attention in revising the BEGs. Moreover, the survey captures only participants' subjective perceptual responses, rather than cognitive or physiological responses, which are more relevant in educational contexts, including public-speaking experiences. Future work should incorporate longer-duration observations and objective, physiological measurements to better assess the impact of BEGs. Following such validation, research can examine BEG-based virtual human manipulation to investigate how emotional gestures influence users' experiences, perceptual responses, and cognitive processes during task performance. Studying how elicited emotions influence participants' experiences and performance could provide insights into mechanisms that support more effective public speaking and potentially extend to the design of educational experiences in VR.

5 CONCLUSION

In this preliminary work, we designed virtual humans' bodily emotional gestures spanning all four quadrants of the valence-arousal space, with representative gestures selected through an online survey ($N = 126$). We discuss implications for emotionally engaging VR environments and offer preliminary design guidelines. Based on these findings, formal evaluation of BEGs in immersive VR is needed to validate their effectiveness and investigate how these emotional gestures influence users' experience, perceptual, and cognitive responses.

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